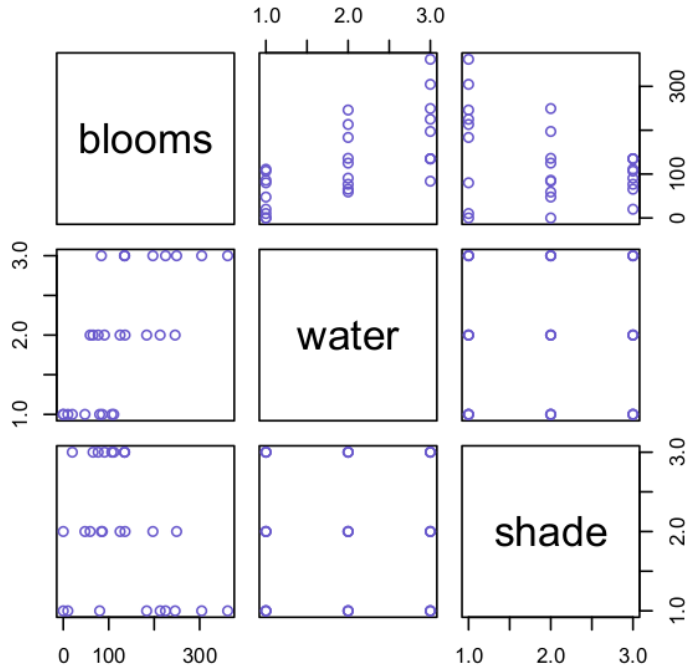
Lecture 24: More interactions

- textttdata(tulips): 27 replicate blooms across three levels of both water and shade.





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With or without interaction

Tulip blooms

- No interaction: water and shade have independent effects

$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_w(w_i - \bar{w}) + \beta_s(s_i - \bar{s})
\end{aligned}
$$

- Interaction: water and shade have interdependent effects



$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_w(w_i - \bar{w}) + \beta_s(s_i - \bar{s}) + \beta_{ws}(w_i - \bar{w})(s_i - \bar{s})
\end{aligned}
$$

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How is a continuous interaction formed?

$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_w(w_i - \bar{w}) + \beta_s(s_i - \bar{s}) + \beta_{ws}(w_i - \bar{w})(s_i - \bar{s})
\end{aligned}
$$

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How is a continuous interaction formed?

$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_w(w_i - \bar{w}) + \beta_s(s_i - \bar{s}) + \beta_{ws}(w_i - \bar{w})(s_i - \bar{s})
\end{aligned}
$$

$$
\begin{aligned}
\mu_i &= \alpha + \gamma_{w,i} W_i + \beta_s S_i \\
\gamma_{w,i} &= \beta_w + \beta_{ws} S_i
\end{aligned}
$$

Tulip model - no interaction

```
m8.4 <- quap(alist(
        blooms_std ~ dnorm(mu , sigma) ,
        mu <- a + bw*water_cent + bs*shade_cent,
        a ~ dnorm(0.5, 0.25),
        bw ~ dnorm(0, 0.25),
        bs ~ dnorm(0, 0.25),
        sigma ~ dexp(1)
    ) , data=d)
```

Plotting a continuous interaction

- The slope changes with values of other predictor(s), so we will use more than one plot.
- Go for "low", "medium" and "high" so we need 3 plots $\rightarrow$ *triptych*

trip•tych | ˈtriptik |

noun

a picture or relief carving on three panels, typically hinged together side by side and used as an altarpiece.

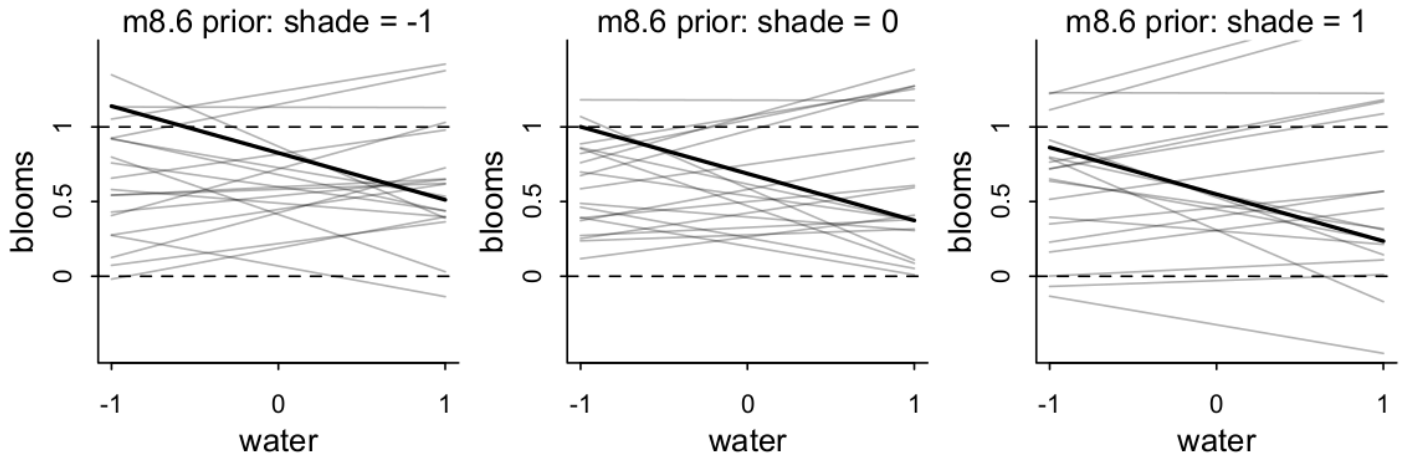
• a set of three associated artistic, literary, or musical works intended to be appreciated together.

ORIGIN mid 18th cent. (denoting a set of three writing tablets hinged or tied together): from TRI- ***'three,'*** on the pattern of *diptych* .



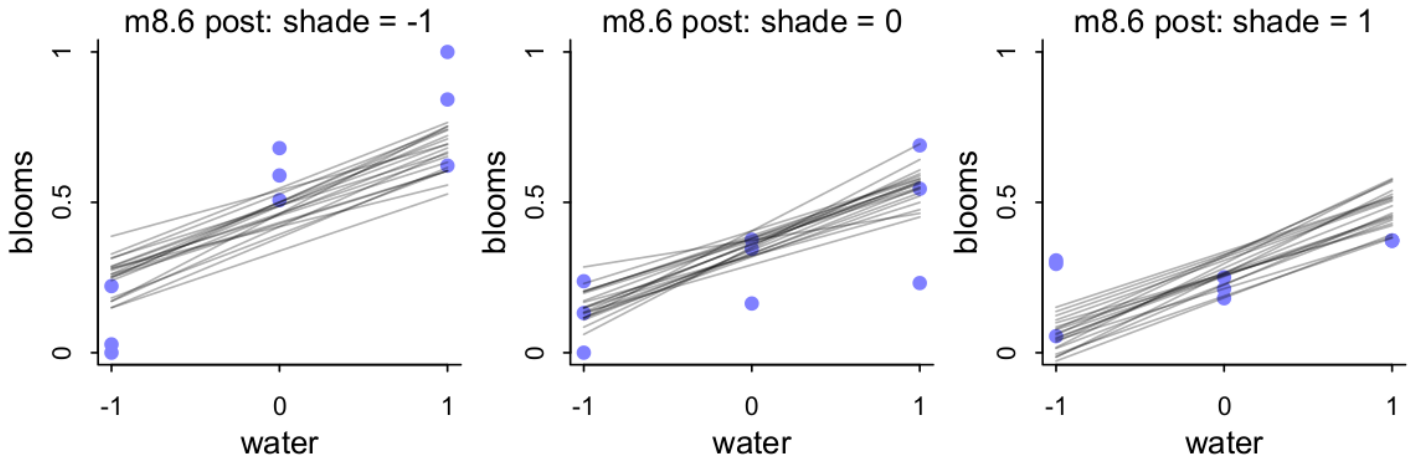
Prior predictions

No interaction



Posterior predictions

No interaction



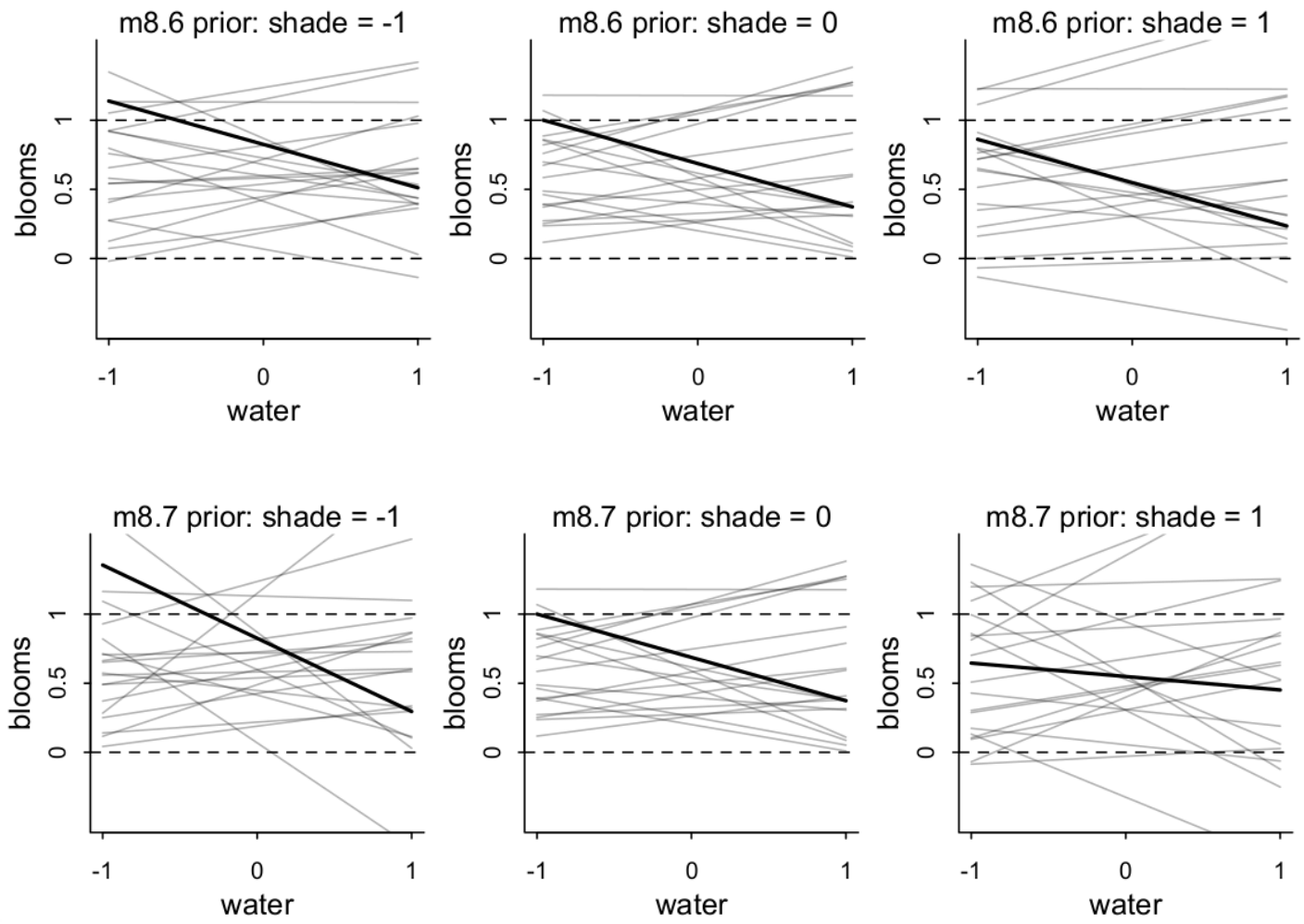
Tulip model - with interaction

```
m8.5 <- quap(alist(
        blooms_std ~ dnorm(mu , sigma) ,
        mu <- a + bw*water_cent + bs*shade_cent +
          bws*water_cent*shade_cent
        a ~ dnorm(0.5, 0.25),
        bw ~ dnorm(0, 0.25),
        bs ~ dnorm(0, 0.25),
        bws ~ dnorm(0, 0.25),
        sigma ~ dexp(1)
    ) , data=d)
```

Since interpreting the respective roles of the parameters is hard, we will plot: use of triptych.

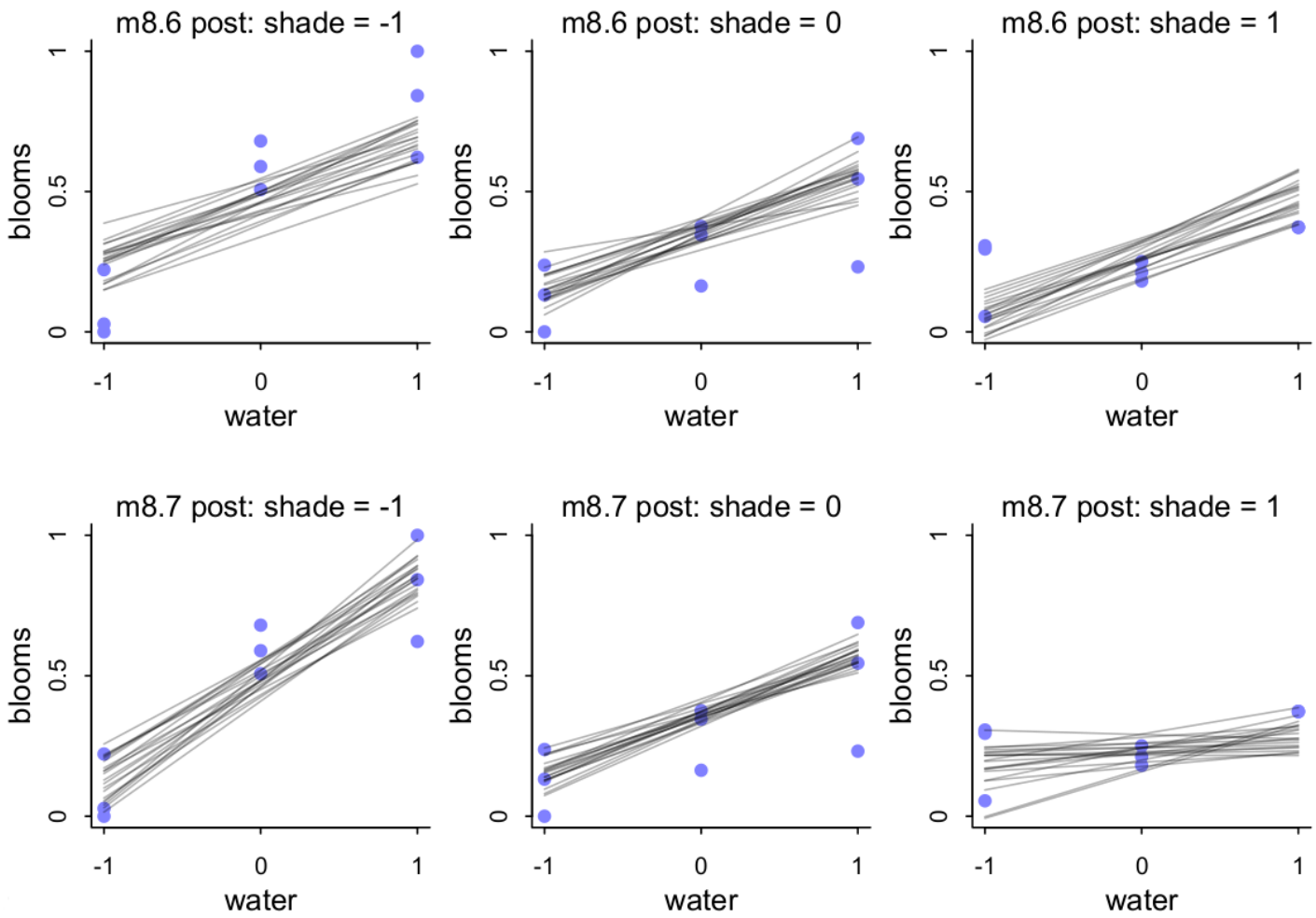
Prior predictions

With interaction (bottom)



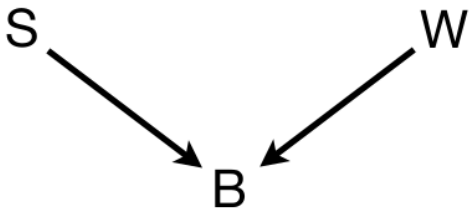
Posterior predictions

With interaction (bottom)

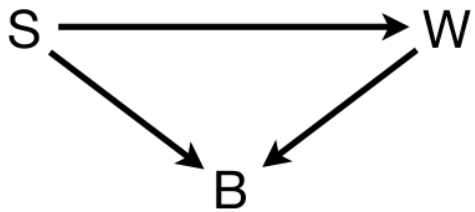


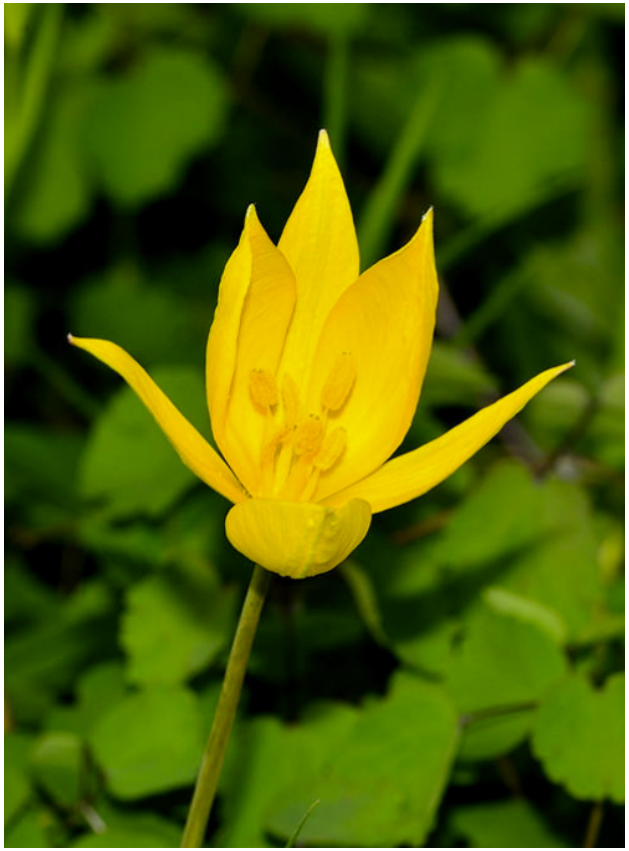
Tulip data - causal thinking

- Tulip experiment:



- Tulip reality:





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Interactions are not always linear

- Suppose all tulip data was collected under "cool" temperatures.
- Under "hot" temperatures, tulips do not bloom.
- Interaction would be observed, but not a linear one
  - Bloom goes to 0 at threshold.



Higher order (linear here) interactions?

- Just keep multiplying

$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma), \\
\mu_i &= \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} \\
&\quad + \beta_{12} x_{1i} x_{2i} + \beta_{13} x_{1i} x_{3i} + \beta_{23} x_{2i} x_{3i} \\
&\quad + \beta_{123} x_{1i} x_{2i} x_{3i}.
\end{aligned}
$$

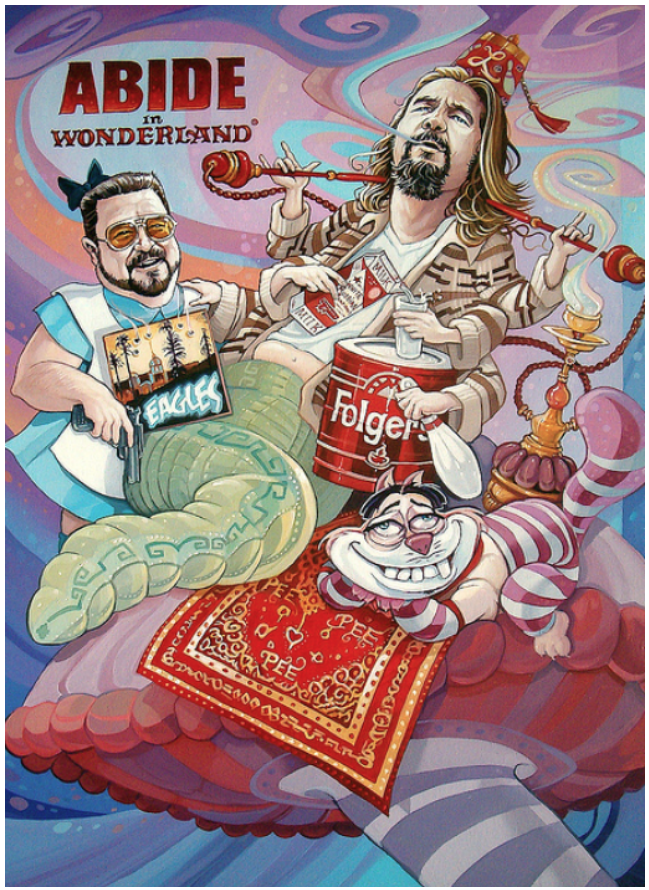
main effects

2-way interactions

3-way interaction

Higher order interactions

- Dangers of higher order interactions
  - Hard to interpret: "The extent to which the effect of x_1 on y depends upon the value of x_2 depends on the value of x_3". Phew!
  - Hard to estimate: needs lots of data. Often need to regularise.
  - But you might really need them, because conditionality runs deep.



Higher order interactions

Wine data example

- `data(Wines2012)`
- Judgment of Princeton, 2012
  - New Jersey wines vs French wines
- Outcome variable: `score`
- Predictors:
  - Region (NJ/FR) (`wine.amer` indicator)
  - Nationality of the judge (USA/FR-BE) (`judge.amer` indicator)
  - `flight` (red/white)



Higher order interactions

Consider the 2nd and 3rd order interactions and their interpretations.

- Interaction of region and judge is bias.
  Bias depends upon flight.
- Interaction of judge and flight is preference.
  Preference depends upon region.
- Interaction of region and flight is comparative advantage.
  Advantage depends on judge

See exercises 8H5, 8H6 and 8H7.



Summary

Over the last two weeks, we have introduced:

- **Overfitting**: models with more parameters fit a sample better. Even when the additional parameters do not make sense. But predictions on out-samples are often worse for over-specified models.
- Regularising priors (reduced overfitting during the estimation) and estimates of out-of-sample accuracy (WAIC and PSIS; they help estimate the degree of overfitting) are tools to address overfitting.
- The tools above do not lead you directly for causal estimation, so model must be designed through another process.
- **Interactions**: the association between a predictor and an outcome depends upon the value of one or several other predictors.
- Cannot be seen in a DAG
- Can be important for accurate inferences.
- Can be difficult to interpret $\rightarrow$ plot the (posterior) predictions (see triptych plots...ore more complex if needed!)

Next week: Markov chains

And then MCMC: Markov chain Monte Carlo



